

MCCLELLAN-PALOMAR, CA, USA

WMO: 722927

Lat: 33.128N

Lon: 117.279W

Elev: 328

StdP: 14.52

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 03177

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	43.0	44.9	11.8	10.2	59.6	17.3	13.3	57.9	17.1	60.9	14.8	59.1	3.1	80	0.337

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.1	84.3	62.9	81.1	64.0	78.5	64.3	71.0	77.5	69.6	75.7	68.7	74.6	6.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.6	106.4	74.3	67.3	101.5	73.5	66.2	97.7	72.7	35.0	77.7	33.9	76.0	33.0	74.8	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.9	10.7	DB	38.5	94.4	1.7	5.1	37.3	98.0	36.3	101.0	35.3	103.8	34.1	107.5	(4)
(5)			WB	32.2	73.0	2.0	1.8	30.8	74.3	29.6	75.3	28.4	76.3	26.9	77.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.5	56.8	56.2	57.7	59.5	62.2	65.0	69.3	70.3	69.3	65.9	60.8	56.1	(6)	
	DBStd	6.62	5.17	4.67	4.23	4.33	3.50	2.92	3.13	3.03	4.23	4.71	4.89	4.30	(7)	
	HDD50	8	4	2	0	0	0	0	0	0	0	0	0	2	(8)	
	HDD65	1523	261	251	232	175	101	33	1	0	3	42	145	279	(9)	
	CDD50	4556	214	176	240	285	378	449	598	628	580	492	325	191	(10)	
	CDD65	594	6	5	6	10	14	32	134	163	133	69	20	2	(11)	
	CDH74	1974	46	42	56	83	96	48	213	336	488	391	154	21	(12)	
	CDH80	422	5	8	12	25	34	8	20	38	117	112	40	3	(13)	
(14) Wind	WSAvg	4.3	3.8	4.5	4.6	5.1	5.3	4.9	4.5	4.1	4.0	3.8	3.6	3.8	(14)	
(15) Precipitation	PrecAvg	11.3	2.3	2.3	1.6	0.9	0.3	0.1	0.1	0.0	0.2	0.6	0.9	1.7	(15)	
	PrecMax	23.0	9.9	10.0	6.3	2.1	1.1	0.8	1.7	0.4	1.1	5.0	3.0	8.0	(16)	
	PrecMin	3.5	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	5.4	2.4	2.2	1.5	0.6	0.3	0.2	0.4	0.1	0.2	1.1	0.8	1.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.1	80.1	80.6	82.7	84.2	79.5	82.2	84.2	89.5	88.3	84.4	76.7	(19)	
		MCWB	55.7	55.6	56.6	56.8	57.8	67.0	69.7	68.6	70.4	62.2	57.7	53.4	(20)	
	2%	DB	73.7	72.7	73.3	74.9	73.7	73.8	78.8	80.5	82.8	82.8	78.4	70.8	(21)	
		MCWB	53.6	53.2	55.0	56.4	60.3	65.4	68.7	68.6	67.7	59.5	56.2	52.4	(22)	
	5%	DB	70.0	67.9	68.1	70.0	69.7	71.6	76.2	77.6	79.3	78.5	73.0	66.7	(23)	
		MCWB	52.7	53.3	54.9	56.5	60.6	64.2	68.1	68.4	68.0	59.9	54.7	52.5	(24)	
	10%	DB	65.7	63.8	64.3	66.0	66.4	69.8	74.1	75.2	75.8	73.5	68.9	63.7	(25)	
		MCWB	52.2	53.2	55.5	57.1	59.2	63.1	67.3	67.7	67.0	61.7	55.4	52.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.1	60.5	61.7	62.2	64.6	67.9	72.2	72.2	72.8	69.8	63.1	61.2	(27)	
		MCDB	66.5	65.3	69.7	70.6	71.9	75.8	78.2	78.9	82.9	75.8	68.3	66.3	(28)	
	2%	WB	58.4	58.4	59.6	60.3	62.6	66.2	70.1	70.4	71.0	67.1	61.3	59.3	(29)	
		MCDB	63.8	63.6	66.0	67.1	68.7	72.7	75.9	76.7	78.4	74.3	67.9	63.0	(30)	
	5%	WB	57.1	57.0	58.3	59.1	61.5	64.9	69.0	69.3	69.5	65.3	60.0	57.7	(31)	
		MCDB	62.3	62.5	64.0	65.3	67.1	70.4	74.6	75.2	75.9	71.7	66.3	61.9	(32)	
	10%	WB	55.7	55.7	57.2	58.1	60.3	63.8	67.9	68.4	68.3	63.9	59.0	56.2	(33)	
		MCDB	61.3	61.5	62.3	63.9	65.2	68.7	73.1	74.1	74.4	69.7	65.0	60.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	14.2	12.3	11.5	9.0	8.2	8.5	9.1	10.6	13.0	15.3	15.6	(35)	
		MCDBR	23.0	21.8	20.0	20.2	14.7	11.3	11.2	12.0	15.7	22.0	22.5	21.2	(36)	
	5% WB	MCWBR	12.3	12.2	11.1	10.7	7.8	5.2	4.6	5.4	7.5	12.6	13.2	13.0	(37)	
		MCWBR	15.4	15.2	13.7	13.8	11.1	11.0	10.4	10.8	12.6	14.8	15.7	14.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.307	0.317	0.334	0.346	0.365	0.362	0.391	0.386	0.357	0.355	0.315	0.317	(40)	
		taud	2.569	2.557	2.493	2.463	2.435	2.449	2.444	2.435	2.500	2.479	2.581	2.574	(41)	
	Ebn at Noon		291	298	300	298	292	291	282	283	288	280	286	281	(42)	
		Edh at Noon	25	29	33	35	36	36	36	36	32	30	25	24	(43)	
(44) All-Sky Solar Radiation	RadAvg		1038	1314	1736	2094	2218	2333	2299	2203	1886	1454	1135	939	(44)	
	RadStd		65	119	93	110	155	145	133	74	87	102	71	68	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+1.67	N/A	N/A	+2.45	+1.71	+1.58	N/A	-378	+551	+238
Regional (45 neighbors)	+1.08	N/A	-2.92	+1.19	N/A	N/A	N/A	-176	+380	+198

Nomenclature: See separate page